University of Florida
College of Public Health & Health Professions

SYLLABUS
PHC 6326: Environmental and One Health
Fall 2025 - 3 credit hours

Delivery Format: In-person (on campus) and E-learning

Room 101 (Auditorium) – Cancer Genomics Research Complex (CGRC)

Thursdays, Periods 3-4 (9:35-11:30am)

Instructor: Andy Kane, MS, PhD
Email Address: kane@ufl.edu
Phone number: 352-213-8407
Office Location and Hours: Office hours by appointment and in classroom after class.
Teaching Assistants: Bianca Punch
Prerequisites: Biological Sciences (BSC 2005), or Environmental Science (EVS 3000), or consent of the instructor

PURPOSE AND OUTCOME

Course Overview

This introductory level course investigates concepts, theories, and applications of environmental health sciences to examine how they contribute to the One Health framework. Worldwide, many public health challenges result from complex and inextricable interactions between humans, animals, and the environment, necessitating a systems approach to addressing these challenges. This One Health approach is utilized to examine sources, routes, media, and health outcomes associated with biological, chemical, and physical agents in the environment. Student (hereafter: you(r)) learning will be facilitated through presentations, activities, assignments, discussions, and a final project.

Similarities and differences between in-person and online course offering

This course closely aligns with the online course offering with regard to learning objectives, curriculum covered, and lectures. However, some assessments, assignments, and specific lectures may differ based on the instructor and online/on-campus differences in course implementation.

Course Learning Outcomes

This course will lead you to:

1. Understand the One Health framework in relation to environmental and public health.
2. Explain how biological, chemical, and physical agents individually and collectively affect disease transmission, water and air quality, food safety, and land resources.
3. Argue a broad range of environmental and public health perspectives from community to occupational settings and developed to developing nations.
4. Interpret major legal frameworks associated with environmental and public health.
5. Relate economic contexts to the main topical areas of the course.

Course Learning Objectives

Upon completion of this course, you will specifically be able to:

1. Describe key concepts and theories of environmental health science and apply the one health systems approach for addressing interdisciplinary public health problems.
2. Define the major sources of biological, chemical, and physical agents found in water, air, soil, and food and describe tools used to analyze health impacts of environmental exposures within such fields as toxicology, exposure assessment, and epidemiology.
3. Examine exposure pathways, disease transmission, toxic mechanisms of action, and health impacts for key toxins and pathogens as related to exemplary human-animal-environmental exposure and disease linkages.
4. Identify major environmental health interventions used to detect, manage, control, or remove health hazards, prevent disease, and improve human and animal health at the individual, community, and population levels.
5. Understand existing regulatory frameworks and policy for controlling environmental and occupational agents/exposures and relate the role of risk assessment paradigms in creating safe environments.
6. Assess the effects of global environmental change on health and describe how both local and global factors affect disease transmission within and between countries.

INSTRUCTIONAL METHODS

Course Delivery and Content

This course is taught as a series of modules, each covering one specific aspect of environmental and one health. Each module may contain lectures, external links, videos, discussions and required readings as well as assignments. You are responsible for all course content regardless of the format. The Topical Outline/Course Schedule below details the dates of content modules, assignments, projects, and exams.

1. Lectures: Lectures are for general orientation. You are responsible for all the material presented in the course and assigned readings.
2. Readings and Resources: Required readings and resources, along with supplementary readings and resources, will be posted in the course site. It is highly recommended that you complete the suggested readings. The reading list may be supplemented during the course.
3. Assessments: A variety of assessments will be used in this course including, but not limited to, assignments, exams, debates, discussions, and a semester project.

Online Faculty Course Evaluation Process

By participating in this course, you are required to complete online evaluations on the quality of course instruction at https://evaluations.ufl.edu. Evaluations are typically open during the last two or three weeks of the semester, but you will be given specific times when they are open. Summary results of these assessments are available to you at https://evaluations.ufl.edu/results/.

Instructor and Learner Agreement

What can you expect from me?

Approachable, reasonable, available, timely with grades/feedback, prepared for class, open to course/assignment feedback, use technology to actively engage.

What is expected of you?

Depending on campus conditions, you are expected to engage with online material, complete all assignments in a timely fashion, and engage with classmates and the instructor through the discussion board.

Course Materials and Technology

Suggested Textbooks

1. One Health: People, Animals, and the Environment. Ronald M. Atlas & Stanley Maloy, ASP Press, 2014 (*OH)*
2. Environmental Health: from Global to Local (3rd edition). Howard Frumkin, Jossey-Bass, Inc., 2016 (*EH*)

In addition to the textbooks, literature for specific lectures are posted within each learning module in Canvas. All textbooks and literature are available electronically through the University of Florida Library. The suggested textbooks are optional, and you can choose to purchase them if you would like this as an extra resource. No exam questions will be taken directly from the book; it is meant to be a supplemental resource.

e-Learning in Canvas site

There will be an online site for this course in Canvas, the learning management system supported by the University. Log in at https://elearning.ufl.edu/ and go to the course site. Here, we will post the syllabus, lecture presentations, modules, assignments, exam reviews, and allow for discussions/chats (e.g. on exams and the project) amongst the students and course leaders. You will also turn in your assignments and class project materials through this site. Once the course begins, all communication will take place through the e-Learning in Canvas site. This includes all emails. This will eliminate any issues with students not getting emails due to connection problems. It will be your responsibility to check the site on a routine basis to keep up with announcements, emails, and course modifications. For technical support related to course materials and links, please contact me, for other technical issues, please contact the UF Help Desk. Their contact information is listed under the SUPPORT SERVICES heading.

Learning Environment

Much of the knowledge content that we would have traditionally presented during a live class lecture is instead provided online. Given the asynchronous nature of the course, It is my goal to set time aside to meet with each student virtually both to get to know you and monitor progress throughout the course.

Public health and health professions are based on the belief in human dignity and on respect for the individual. As we share our personal beliefs inside or outside of the classroom, it is always with the understanding that we value and respect diversity of background, experience, and opinion, where every individual feels valued. We believe in, and promote, openness and tolerance of differences in ethnicity and culture, and we respect differing personal, spiritual, religious, and political values. We further believe that celebrating such diversity enriches the quality of the educational experiences we provide our students with and enhances our own personal and professional relationships. We embrace The University of Florida's Non-Discrimination Policy, which reads, "The University shall actively promote equal opportunity policies and practices conforming to laws against discrimination. The University is committed to non-discrimination with respect to race, creed, color, religion, age, disability, sex, sexual orientation, gender identity and expression, marital status, national origin, political opinions or affiliations, genetic information, and veteran status as protected under the Vietnam Era Veterans' Readjustment Assistance Act." If you have questions or concerns about your rights and responsibilities for participating in an inclusive learning environment, please see your instructor or refer to the Office of Multicultural & Diversity Affairs website: www.multicultural.ufl.edu

Topical Outline/Course Schedule

The course instructor reserves the right to modify the schedule as necessary.

Getting Started

1. Visit http://lss.at.ufl.edu and login to e-Learning in Canvas using your Gatorlink ID and password.
2. Find our course website. It will be listed as PHC6326: Environmental and One Health, Fall 2025.

3. Complete the "Getting Started" Module under the Modules Tool (left menu). This will prompt you to download and review the syllabus, review the materials on plagiarism, and complete the syllabus quiz.

The remainder of the course materials will be locked and unavailable to you until you have completed the "Getting Started" module. You MUST earn a 100% score on this quiz for the course materials to open on the course site. If you do not receive a 100% score, please review the feedback on your quiz attempt and retake it as soon as possible. This is an important element to ensure that all students are aware of the curriculum requirements for this course. If you have ANY difficulty with this quiz, please send an email in the course using the Canvas "Inbox" as soon as possible.

Tentative schedule

Week/Module Meeting Dates	Topic(s)	In the News Presentations	Assignments	Class Activities
Getting Started **Week 1** 8/21	Syllabus, course expectations, Introduction to Environmental and One Health	-	Syllabus Quiz	Course structure and expectations, introductions
Week 2 8/28	One Health Systems Thinking	-	Assignment 1 due	Assignment - Solving a Mystery using the One Health Concept
Week 3 9/4	Environmental Epidemiology	Environmental Epidemiology ITN		Environmental Epidemiology ITN follow-up discussion
Week 4 9/11	Toxicology and Environmental Disease; Emerging Contaminants	Toxicology ITN	Assignment 2 due	Assignment - Toxicants as emerging threats Toxicology ITN follow-up discussion
Week 5 9/18	Environmental Infectious Disease	Environmental Infectious Disease ITN		Environmental Infectious Disease ITN follow-up discussion
Week 6 9/25	Emerging Zoonotic Diseases and Global Health Security	Emerging Zoonotic Diseases and Global Health Security ITN	Assignment 3	Emerging Zoonotic Diseases and Global Health Security ITN follow-up discussion
Week 7 10/2	**MIDTERM EXAM**			
Week 8 10/9	Food Quality and Health	Food Quality and Health ITN		Food Quality and Health ITN follow-up discussion
Week 9 10/16	Air Quality and Health	Air Quality ITN		Air Quality ITN follow-up discussion
Week 10 10/22	Water Quality and Health	Water Quality and Health (WASH) ITN		WASH ITN follow-up discussion
Week 11 10/30	Climate Change & Global Health	Climate Change and Global Health ITN	Assignment 4 due	Assignment 4 – Using a One Health Approach to Forecast Coastal Environmental Change Climate Change and Global Health ITN follow-up discussion

Week 12	Environmental monitoring and exposure			Environmental Monitoring ITN follow-up discussion
Week 13	Risk Assessment, Management & Communication	Risk Assessment, Management & Communication ITN		Risk Assessment, Management & Communication ITN Follow-up discussion
Week 14	Environmental Justice & Health Disparities	Environmental Justice & Health Disparities ITN	Assignment 5 due	Environmental Justice & Health Disparities ITN Follow-up discussion
Week 15	Holiday break – No classes			
Week 16	**FINAL EXAM**			

Grading

The course assignments/activities are broken down into the following points: Requirement	**Due date**	**Points**
Written Assignments = 5 @ 60 pts each	See Course Schedule	300
In the News - lead presentation (1)	See Course Schedule and assignment	100
In the News - lead write up (1)	See Course Schedule and assignment	100
In the News Reflections (10 @ 10 pts each)	See Course Schedule and topic assignment	100
Midterm Exam = 1 @ 200 points	See Course Schedule	200
Final Exam = 1 @ 200 points	See Course Schedule	200
	TOTAL POINTS:	**1,000**

You will be graded in the course through the use of different learning assessments which are as follows:

Written Assignments (5 @ 60 points each; Total 300 points)

Purpose and relationship to course goals: The written assignments are an opportunity for students to apply their knowledge gained in the readings and lectures. Each assignment takes a different approach to a selection of the modules with engaging activities, ranging from analyzing and interpreting data to drafting a grant proposal idea.

There will be five (5) graded written assignments. Typically, these will be one to two single-spaced pages long. Detailed instructions will be provided with each assignment. Page limits and content limits do not include reference lists unless otherwise noted.

Assignments 1 - 5 are to be turned in as a text entry or Word document in Canvas by the date assigned unless otherwise indicated. They will be returned back to you with comments after grading, as applicable. If you have unexpected issues with Canvas, you may email the assignment to the course TA and professor directly and submit it on Canvas once the issues are resolved. All written assignments are individual work, unless otherwise directed. Shared work (other than shared efforts on team projects) will be treated as a form of plagiarism. Assignments may be required to be submitted via Turnitin in this course (this will be done automatically in the Canvas Assignment). This tool will pick up any passages in students' work that come from another source. Be sure to adequately cite your sources/references for these assignments to avoid plagiarism (see format below). Please check your Turnitin report by going back to the assignment and clicking on the colored box icon in the assignment (you want to achieve a blue or green indication) – anything yellow, orange or red should be

adjusted and resubmitted BEFORE the assignment deadline. So, it is recommended that you submit EARLY to enable you to utilize this option of resubmission. Resubmissions after the due date are subject to a late submission penalty.

In the News Assignments (3 @ 100 points each = 300 total points)

Purpose and relationship to course goals: The 'In the News' presentations connect students with course connect, current events, and current conversations in the media. Students apply critical thinking skills and practice oral and written communication skills.

Each student will be required to submit and lead one 'In the News' assignment during the semester. For this assignment, each student will be assigned to a topic area at the beginning of the semester, such as 'toxicology' or 'infectious disease.' You will perform a search to identify a current news story that relates to this topic. The 'news' can be positive, negative, changes in policy, public concerns, etc. You may use public news forums (TV, web, newspaper); however, please make sure that the sources are credible. If you need help determining whether a source is credible or suitable for use in this assignment, please reach out to your course instructor or teaching assistant, as applicable.

There are two components associated with each presentation:

(1) You are to prepare 1 video presentation (5 minutes maximum) that will be posted to canvas (100 points). Each student will develop a 5-7 minute PowerPoint presentation that makes the topic easily understood by a general audience, while covering the content credibly, effectively, and creatively. The presentation must follow assignment format and content. (see "In the news" assignment description on Canvas).

- Describe the 'News' item in brief and how it relates to the assigned category (i.e., toxicology).
- Is the news projecting or emphasizing a positive influence on public health or a concern? What evidence do they have?
- What do the experts say should be done about it? Is there any controversy?

(2) For your assigned ITN lead presentation, you are to prepare a one-page description of the article including response points as described in the assignment in Canvas. Both the PowerPoint document and the one-page description must be uploaded into Canvas by 5pm on the Tuesday prior to your presentation date. (100 points).

a. Outline the strengths and weaknesses of the article you selected (example topics to consider include descriptions of scientific evidence, justifications, coverage of the topic, expectations/knowledge, and similar). (40pts)
b. Comment on the communication approach used by this article – who do you think is the target audience, and how could this be improved. (40pts)
c. Were you familiar with this topic prior to your ITN? (10pts)
d. Be sure to cite all sources. (10pts)

An example 'In the News' presentation will be posted on Canvas. Make sure to cite your references. The due date for this assignment will be based on the topic you are assigned as due dates are associated with the specific topic.

Reflections for other's In The News presentations (100 points). All students (except those assigned to make ITN presentations that week) are expected develop a reflection on ITNs presented in class. Reflections should

include all points indicated in the Reflections assignment (provided in Canvas). **(10 ITN reflections @10 points each).**

All ITN reflections must be uploaded into Canvas no later than 11:59 PM the day following the ITN presentations in class.

Exams (one midterm and one final @ 200 points each; Total 400 points)
Purpose and relationship to course goals: The exams are an opportunity for students to demonstrate course content and knowledge, apply their knowledge to new situations, show their understanding of the connections among course themes through a variety of question styles and types, and apply and reinforce course materials.

The format for both exams will be **CLOSED BOOK**. The midterm exam will test your knowledge of the first series of modules, including material covered in lectures, *In The News* presentations, and assigned readings. The final exam will focus on material covered in the second half of the course. Since materials in the second half of the course builds on concepts presented during the first half of the course it is important to have a good comprehension of material covered during the first part of the course (i.e., the final exam is cumulative). Both exams will be comprised of multiple choice, matching, and true/false questions, where you will be expected to interpret the material that is presented in the course and apply it to the scenarios or situations that the exam questions present.

Students will take the exams in class (during class period) using a Lockdown Browser.

ACADEMIC REQUIREMENTS
General information
Assignments are normally intended as individual projects unless otherwise directed. Shared work may be treated as a form of plagiarism. **Do your best to write content from outside sources into your own words and cite the source using an in-text citation.** The Canvas assignment tool may notify you confirming the submission of your assignment. PLEASE check your UFL email at http://webmail.ufl.edu on a regular basis for these and other email notices from the course site. If you do not receive an email confirmation within 2 hours of submission, please return to the site and resubmit your assignment. It is your responsibility to verify that you turn in assignments on time and that you turn in the CORRECT assignment. Please take a few moments to open your submitted attachment and verify that you have submitted the correct file.

Point system used (i.e., how do course points translate into letter grades):

Points earned	93-100%	90-92%	87-89%	83-86%	80-82%	77-79%	73-76%	70-72%	67-69%	63-66%	60-62%	Below 62%
Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E

Please be aware that a C- is not an acceptable grade for graduate students. In addition, a grade of C counts toward a graduate degree only if an equal number of credits in courses numbered 5000 or higher have been earned with an A.

For greater detail on the meaning of letter grades and university policies related to them, see the Registrar's Grade Policy regulations at:
http://catalog.ufl.edu/ugrad/current/regulations/info/grades.aspx

SUPPORT SERVICES

Technical support
UF Help Desk:
Available 24 hours a day, 7 days a week
- (352) 392-HELP - select option 2
- helpdesk@ufl.edu (email)
- helpdesk.ufl.edu (website)

Academic Resources
• Career Connections Center: Reitz Union Suite 1300, 352-392-1601. Career assistance and counseling services.
• Library Support: Various ways to receive assistance with respect to using libraries or finding resources.
• Teaching Center: Broward Hall, 352-392-2010 or to make an appointment 352- 392-6420. General study skills and tutoring.
• Writing Studio*:* 2215 Turlington Hall*,* 352-846-1138. Help brainstorming, formatting, and writing papers.
• Student Complaints On-Campus: Visit the Student Honor Code and Student Conduct Code webpage for more information.
• On-Line Students Complaints: View the Distance Learning Student Complaint Process.

Accommodations for Students with Disabilities
If you require classroom accommodation because of a disability, you must register with the Dean of Students Office http://www.dso.ufl.edu within the first week of class. The Dean of the Student's Office will provide documentation of accommodations to you, which you will then give to me as the instructor of the course to receive accommodations. Please make sure you provide this letter to me by the end of the second week of the course. The College is committed to providing reasonable accommodations to assist students in their coursework.

Counseling and Student Health
Students sometimes experience stress from academic expectations and/or personal and interpersonal issues that may interfere with their academic performance. If you find yourself facing issues that have the potential to or are already negatively affecting your coursework, you are encouraged to talk with an instructor and/or seek help through University resources available to you.

• The Counseling and Wellness Center 352-392-1575 offers a variety of support services such as psychological assessment and intervention and assistance for math and test anxiety. Visit their web site for more information: http://www.counseling.ufl.edu. Online and in person assistance is available.

• You Matter We Care website: http://www.umatter.ufl.edu/. If you are feeling overwhelmed or stressed, you can reach out for help through the You Matter We Care website, which is staffed by Dean of Students and Counseling Center personnel.

• The Student Health Care Center at Shands is a satellite clinic of the main Student Health Care Center located on Fletcher Drive on campus. Student Health at Shands offers a variety of clinical services. The clinic is located on the second floor of the Dental Tower in the Health Science Center. For more information, contact the clinic at 392-0627 or check out the web site at: https://shcc.ufl.edu/

• Crisis intervention is always available 24/7 from:
Alachua County Crisis Center (352) 264-6789
http://www.alachuacounty.us/DEPTS/CSS/CRISISCENTER/Pages/CrisisCenter.aspx

Do not wait until you reach a crisis to come in and talk with us. We have helped many students through stressful situations impacting their academic performance. You are not alone so do not be afraid to ask for assistance.

NECESSARY CONTRACTUAL STUFF

Behavior
You are expected to maintain a civil tone and respect the opinions of others. While commenting on others' statements and ideas is encouraged, aggressive or patronizing tone and language are unacceptable and may result in the loss of your posting and discussion privileges. This type of unacceptable behavior and penalty may impact course and assignment grades, as applicable.

Use of AI
Use of AI is strictly prohibited in this course, unless specifically authorized in an assignment. Our goal is to train you to be independent critical thinkers, which requires a effort and commitment to completing the work on your own without the assistance of AI.

Late or Makeup Assignments
Assignments turned in up to 24 hours late will automatically be discounted 10% of the grade that they would otherwise receive. Assignments turned in more than 24 hours late will not be graded and will contribute zero points toward your final grade unless arrangements have been made in advance with the instructor or in the case of extenuating circumstances. Extenuating circumstances will be addressed on a case-by-case basis. Missed assignments will contribute zero points toward your final grade.

Special Circumstances
In the event of exceptional situations that may interfere with your ability to perform an assignment or meet a deadline, contact the instructor as soon in advance of the deadline as possible. Such special cases will be dealt on an individual basis, provided that you have sufficient documentation. Please note: Any requests for make-ups due to *technical issues* MUST be accompanied by the ticket number received from LSS when the problem was reported to them. The ticket number will document the time and date of the problem. You MUST e-mail me within 24 hours of the technical difficulty if you wish to request a make-up.

Privacy
As in all courses, unauthorized recording and unauthorized sharing of recorded materials is prohibited. Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor.

Publication without permission of the instructor is prohibited. To "publish" means to share, transmit, circulate, distribute, or provide access to a recording, regardless of format or medium, to another person (or persons), including but not limited to another student within the same class section. Additionally, a recording, or transcript of a recording, is considered published if it is posted on or uploaded to, in whole or in part, any media platform, including but not limited to social media, book, magazine, newspaper, leaflet, or third party

note/tutoring services. A student who publishes a recording without written consent may be subject to a civil cause of action instituted by a person injured by the publication and/or discipline under UF Regulation 4.040 Student Honor Code and Student Conduct Code.

Communication Guidelines

Students should communicate with the instructor and the TA by email using the "Inbox" in Canvas for clarification and assistance with the course material and the assignments, and for special issues that may arise. Weekday daytime (US Eastern Time) emails have the best chances of being answered quickly. Please only use the Canvas "Inbox" to communicate with the course instructor and/or TA. While the instructors and TAs will check their UFL email regularly, they will not be held responsible for email sent directly to their UFL addresses. The goal is to receive a reply within 48 hours during the work week (Monday through Friday) or a reply to a weekend email after the next week begins. Please do not anticipate replies to email during the weekend.

Academic Integrity

Students are expected to act in accordance with the University of Florida policy on academic integrity. As a student at the University of Florida, you have committed yourself to uphold the Honor Code, which includes the following pledge:

"**We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honesty and integrity**."

You are expected to exhibit behavior consistent with this commitment to the UF academic community, and on all work submitted for credit at the University of Florida, the following pledge is either required or implied:

"On my honor, I have neither given nor received unauthorized aid in doing this assignment."

It is your individual responsibility to know and comply with all university policies and procedures regarding academic integrity and the Student Honor Code. Violations of the Honor Code at the University of Florida will not be tolerated. Violations will be reported to the Dean of Students Office for consideration of disciplinary action. For additional information regarding Academic Integrity, please see Student Conduct and Honor Code or the Graduate Student Website for additional details:

https://www.dso.ufl.edu/sccr/process/student-conduct-honor-code/
http://gradschool.ufl.edu/students/introduction.html

Please remember cheating, lying, misrepresentation, or plagiarism in any form is unacceptable and inexcusable behavior.

Plagiarism is especially damaging in an online learning environment and will be dealt with in an official way, according to University of Florida regulations. Incidents will be reported directly to the Office of Student Judicial Affairs and a formal procedure will be started in each and every case. There will be no informal incident resolution between student and instructor. Should you have any doubts on whether something constitutes plagiarism, please consult the many available resources on the topic, e.g., starting with http://web.uflib.ufl.edu/msl/subjects/Physics/StudentPlagiarism.html, or contact the instructor in advance. There is also a reference posted on the course site, in the Course Help link. As you submit assignments, you will have the opportunity to check it for unintentional plagiarism using Turnitin®, the same software that instructors will use to check your work. You are encouraged to take advantage of this option. If you turn in assignments that are plagiarized, you will receive zero points for that assignment.

You are expected to turn in original work in this course. This means that when answering assignment questions, writing papers, posting discussions, etc., you will be expected to write your responses in your own words. You **MAY NOT** copy answers word for word from any course materials or outside sources. On occasion it may be useful to provide a quote from course materials or outside sources in which case you must properly

cite the source, **in APA format**, and place the quote in quotation marks. That being said we urge you to avoid excessive quotation as it does little to demonstrate your understanding of course material.